

Year 11
Mathematics Extension 1
Assessment Task 3
Yearly Examination
2012

General Instructions

- Reading time – 5 minutes
- Working time – 1.5 hours
- Write using black or blue pen
- Board-approved calculators may be used
- All necessary working should be shown in every question

Note: Any time you have remaining should be spent revising your answers.

Total marks – 55

Section I

10 marks

- Attempt Questions 1 – 10

Section II

45 marks

- Attempt Questions 11, 12 & 13
- Start each question in a new writing booklet
- Write your name on the front cover of each booklet and writing sheet to be handed in.
- If you do not attempt a question, submit a blank booklet marked with your name and "N/A" on the front cover

DO NOT REMOVE THIS PAPER FROM THE EXAMINATION ROOM

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Section A

10 marks

Attempt Questions 1 - 10

Use the multiple-choice answer sheet.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☐ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
 correct
 ↙

Section I**10 marks****Attempt Questions 1 – 10**

Use the multiple-choice answer sheet for Questions 1-10

- 1 How many different nine letter combinations can be made using the letters forming the word **AUSTRALIA**?
(A) 720
(B) 60 480
(C) 362 874
(D) 362 880

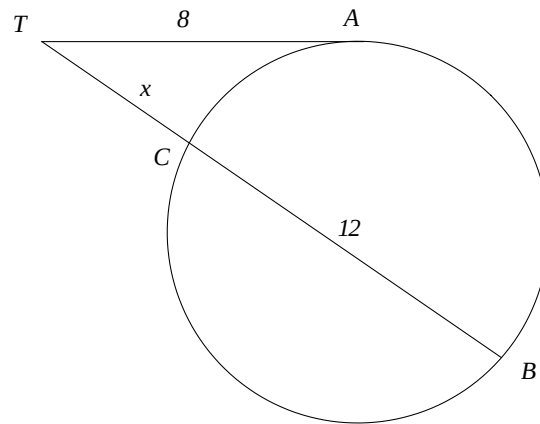
- 2 What is the acute angle between the lines $2x - y - 7 = 0$ and $3x - 5y - 2 = 0$?
(A) $4^{\circ}24'$
(B) $85^{\circ}36'$
(C) $57^{\circ}32'$
(D) $32^{\circ}28'$

- 3 Which of the following is equal to $\frac{\sin(90 - \theta)}{\cos(180 + \theta)}$?
(A) -1
(B) $-\frac{1}{2}$
(C) $\frac{1}{2}$
(D) 1

- 4 State the largest possible domain of the function:

$$f(x) = \sqrt{x+3} - \sqrt{x-3}$$

- (A) $x \leq -3, x \geq 3$
- (B) $x \geq 3$
- (C) $-3 \leq x \leq 3$
- (D) $x \geq -3, x \geq 3$
- 5 The line TA is a tangent to the circle at A and TB is a secant meeting the circle at B and C .

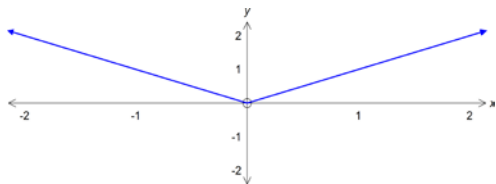


Given that $TA=8$, $CB=12$ and $TC=x$, what is the value of x ?

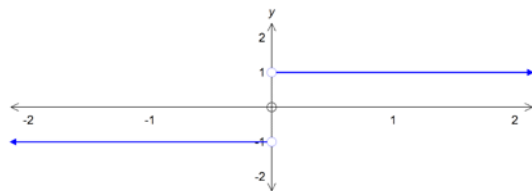
- (A) 16
- (B) $4\frac{3}{8}$
- (C) 4
- (D) $6\frac{2}{3}$

6 The sketch of the graph of $f(x) = \frac{|x|}{x}$, $x \neq 0$ is:

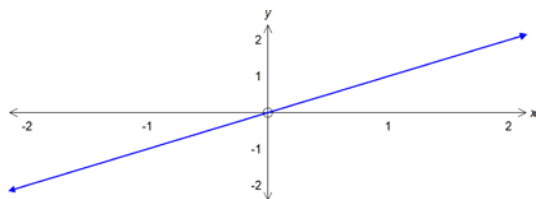
(A)



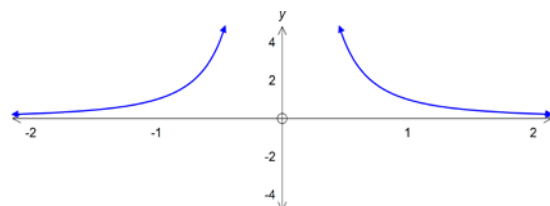
(B)



(C)



(D)



7 Given that $\sin \alpha = \frac{1}{4}$, what is the value of $\cos 2\alpha$?

(A) $\frac{1}{2}$

(B) $\frac{3}{4}$

(C) $\frac{7}{8}$

(D) $\frac{2\sqrt{15}}{16}$

8 In how many ways can the letters of the word RHOMBUS be arranged so that the vowels are together?

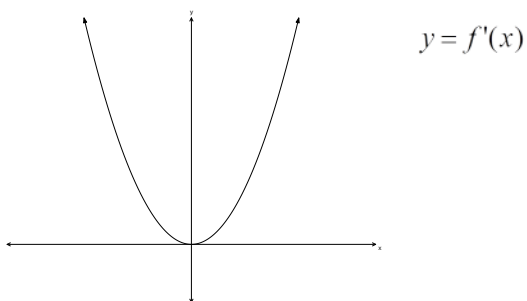
(A) 720

(B) 5040

(C) 1440

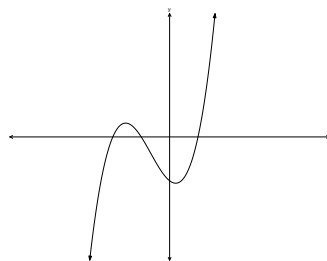
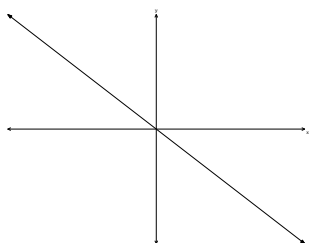
(D) 360

9

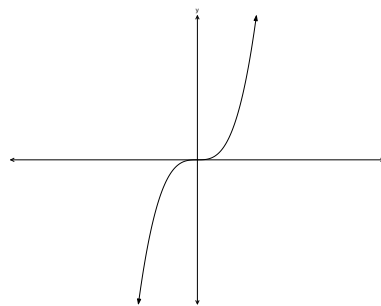
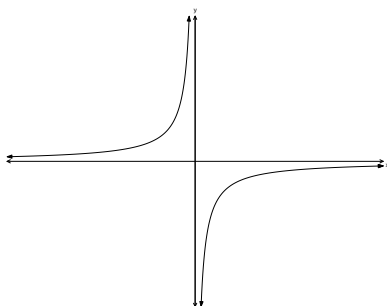


Given the graph of $y = f'(x)$ above, which graph best represents $y = f(x)$?

(A) (B)



(C) (D)



10 Evaluate $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2}$

- (A) 0
- (B) undefined
- (C) 4
- (D) 12

END OF SECTION I

Section II**45 marks****Attempt Questions 11 – 13**

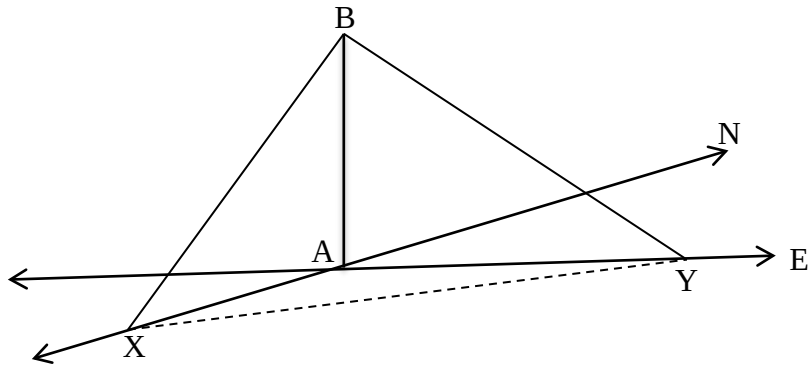
Answer each question in the appropriate writing booklet.

All necessary working should be shown in every question.

Question 11 (15 Marks)		Marks
a)	Solve for x : $16^{4-x} = \frac{1}{8^x}$	2
b)	Find the coordinates of the point P which divides the interval AB joining $A(1, -3)$ and $B(2, -2)$ externally in the ratio $2:3$.	2
c)	How many different committees of 8 people with equal numbers of men and women can be chosen if there are 6 women and 10 men?	2
d)	i) Find the x coordinate of the point of intersection of $y = x^2 - 2x$ and $y = x^2 - 7x + 10$.	1
	ii) Hence find the acute angle between these two curves at that point. Give your answer to the nearest degree.	2
e)	Solve for x : $\frac{x^2 - 4}{x} > 0$.	3

Question 11 continues on the next page.

f)

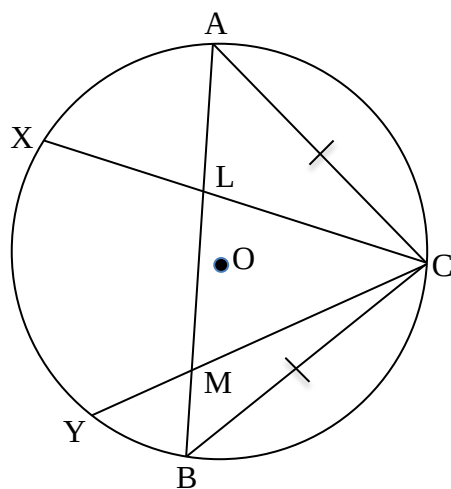


A pole AB is 4 metres high and on level ground. Point X is due south of the pole and the angle of elevation from X to the top of the pole is 45° . Another point Y is due east of the pole and the angle of elevation from Y to the top of the pole is 30° .

- i) Find the exact value for AX and AY . **1**
- ii) Find the distance XY . **2**

Question 12 (15 Marks)**Marks**

a)



In the diagram A, B and C are three points on the circle. Chords CX and CY cut AB at L and M respectively. $AC = CB$, $\angle CAB = \theta$ and $\angle ACX = \alpha$.

Copy or trace the diagram onto your paper.

- | | | |
|------|---|----------|
| i) | State why $\angle CLB = \theta + \alpha$. | 1 |
| ii) | Explain why $\angle AYC = \theta$ and $\angle AYX = \alpha$. | 2 |
| iii) | Prove that $XYML$ is a cyclic quadrilateral. | 1 |

- b) Find the derivative of the following functions, giving your answer in fully factorised form where appropriate.

i)	$y = \frac{3x^2 - 2x + 1}{\sqrt{x}}$	2
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ii)	$y = \frac{x^2}{1 - x^3}$	2
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iii)	$y = (x + 1)\sqrt{4x - 1}$. For this question you must give your answer in fully factorised form.	3
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Question 12 continues on the next page.

- c) The tangent to $y = \frac{a}{x+2}$ at $x = 2$ has a gradient of $\frac{1}{4}$. **2**
Find the value of a .
- d) One of the roots of $x^2 + bx + c = 0$ is 3 times the other root. **2**
Show that $3b^2 - 16c = 0$

Question 13 (15 Marks)**Marks**

- a) If $t = \tan \frac{\theta}{2}$ show that: $\frac{\cos \theta}{1 - \sin \theta} = \frac{1+t}{1-t}$. **3**
- b) i) Express $\sqrt{3} \sin x - \cos x$ in the form $R \sin(x - \alpha)$, where $R > 0$ and $0^\circ \leq \alpha \leq 90^\circ$. **2**
- ii) Hence or otherwise solve $\sqrt{3} \sin x - \cos x = 1$ for $0^\circ \leq x \leq 360^\circ$. **2**
- c) i) Show that $\frac{1 - \cos 2\theta}{1 + \cos 2\theta} = \tan^2 \theta$ **2**
- ii) Hence or otherwise, show that the exact value of $\tan 22\frac{1}{2}^\circ = \sqrt{2} - 1$. **2**
- d) i) Find any values of a and b such that $x^4 + 4 \equiv (x^2 + a)^2 - (bx)^2$. **2**
- ii) Hence express $x^4 + 4$ as a product of two quadratic factors. **2**

END OF EXAMINATION

Preliminary Yearly Examination – Extension 1 Mathematics 2012
Multiple Choice Answer Sheet

Name _____

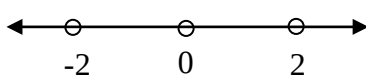
Completely fill the response oval representing the most correct answer.

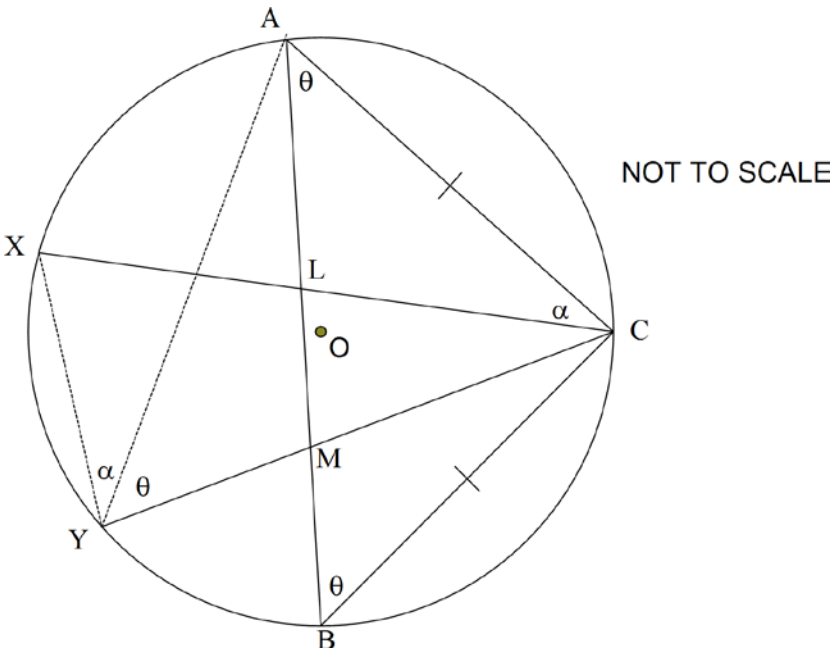
1. A ○ B ○ C ○ D ○
2. A ○ B ○ C ○ D ○
3. A ○ B ○ C ○ D ○
4. A ○ B ○ C ○ D ○
5. A ○ B ○ C ○ D ○
6. A ○ B ○ C ○ D ○
7. A ○ B ○ C ○ D ○
8. A ○ B ○ C ○ D ○
9. A ○ B ○ C ○ D ○
10. A ○ B ○ C ○ D ○

MULTIPLE CHOICE

1. B
2. D
3. A
4. B
5. C
6. B
7. C
8. C
9. D
10. D

Question 11		Preliminary Mathematics Extension 1	2012	
Part	Solution	Marks	Comment	
(a)	$16^{4-x} = \frac{1}{8^x}$ $(2^4)^{4-x} = 2^{-3x}$ $2^{16-4x} = 2^{-3x}$ $16 - 4x = -3x$ $x = 16$	2	1 1	
(b)	$A(1, -3)$ and $B(2, -2)$, Externally 2 : -3 $x = \frac{mx_2 + nx_1}{m + n}$ $= \frac{2 \times 2 + -3 \times 1}{2 + -3}$ $= -1$ $y = \frac{my_2 + ny_1}{m + n}$ $= \frac{2 \times -2 + -3 \times -3}{2 + -3}$ $= -5$ The coordinates of P are $(-1, -5)$	2	1 Correct use of formula 1 correct coordinates	
(c)	${}^6C_4 \times {}^{10}C_4 = 3150$	2	1 mark for correct combinations 1 mark correct numerical answer	
(d)	i) For intersection of $y = x^2 - 2x$ and $y = x^2 - 7x + 10$: $x^2 - 2x = x^2 - 7x + 10$ $5x = 10$ $x = 2$ ii) For $y = x^2 - 2x$, $y' = 2x - 2$ then $m_1 = 2$ For $y = x^2 - 7x + 10$, $y' = 2x - 7$ then $m_2 = -3$ $\tan \theta = \left \frac{m_1 - m_2}{1 + m_1 m_2} \right $ $= \left \frac{2 - (-3)}{1 + 2 \times (-3)} \right $ $= \left \frac{5}{-5} \right $ $= 1$ $\theta = 45^\circ$	1 2	1 mark correct 1 correct substitution into correct formula 1 correct answer	

Question 11		Preliminary Mathematics Extension 1	2012	
Part	Solution		Marks	Comment
(e)	$\frac{x^2 - 4}{x} > 0$ $\frac{x^2(x^2 - 4)}{x} > 0 \times x^2$ $x(x - 2)(x + 2) > 0$ $-2 < x < 0, x > 2$ OR $\frac{x^2 - 4}{x} > 0, x \neq 0$ $(x + 2)(x - 2) = 0$ $x = \pm 2$ Test: $x = 1, \frac{1 - 4}{1} = -3 < 0$ $x = -1, \frac{1 - 4}{-1} = 3 > 0$ So $-2 < x < 0, x > 2$ 		3	1 mark any correct working 1 mark each for correct inequalities
(f)(i)	In $\triangle ABX$ $\tan 45^\circ = \frac{4}{AX}$ $AX = \frac{4}{\tan 45^\circ}$ $= 4$	In $\triangle ABE$ $\tan 30^\circ = \frac{4}{AY}$ $AY = \frac{4}{\tan 30^\circ}$ $= \frac{4}{\frac{1}{\sqrt{3}}} = 4\sqrt{3}$	1	1 mark correct
(ii)	Pythagoras theorem $XY^2 = BX^2 + BY^2$ $= 4^2 + (4\sqrt{3})^2$ $= 64$ $XY = 8 \text{ cm}$		2	1 mark recognises Pythagoras' theorem (or may for some reason use cosine rule) 1 mark correct answer

Question 12		Preliminary Mathematics Extension 1	2012	
Part	Solution		Marks	Comment
(a)				
(i)	The exterior angle of a triangle ($\triangle ALC$) is equal to the sum of the opposite interior angles.			1
(ii)	$\angle ABC = \angle BAC = \theta$ (base angles of isos $\triangle ACB$) $\angle AYC = \angle ABC = \theta$ (angles standing on the same arc are equal) $\angle AXY = \angle ACX = \alpha$ (angles standing on the same arc are equal)		2	1 mark each
(iii)	$\angle XYM = \alpha + \theta$ (all shown above) $= \angle CLB$ $= \alpha + \theta \therefore YMLX$ is a cyclic quadrilateral. (exterior angle of cyclic quad is equal to opp interior angle)			1

Question 12		Preliminary Mathematics Extension 1	2012	
Part	Solution	Marks	Comment	
(b)i)	$y = \frac{3x^2 - 2x + 1}{\sqrt{x}} = 3x^{\frac{3}{2}} - 2x^{\frac{1}{2}} + x^{-\frac{1}{2}}$ $y' = 3 \times \frac{3}{2} \times x^{\frac{1}{2}} - 2 \times \frac{1}{2} \times x^{-\frac{1}{2}} + -\frac{1}{2} x^{-\frac{3}{2}}$ $= \frac{9}{2} x^{\frac{1}{2}} - x^{-\frac{1}{2}} - \frac{1}{2} x^{-\frac{3}{2}} \quad \text{OR}$ $= \frac{9}{2} \sqrt{x} - \frac{1}{\sqrt{x}} - \frac{1}{2x\sqrt{x}}$ OR (and this is much messier) $u = 3x^2 - 2x + 1 \quad v = x^{\frac{1}{2}}$ $u' = 6x - 2 \quad v' = \frac{1}{2} x^{-\frac{1}{2}}$ $y' = \frac{x^{\frac{1}{2}}(6x - 2) - \frac{1}{2} x^{-\frac{1}{2}}(3x^2 - 2x + 1)}{\left(x^{\frac{1}{2}}\right)^2}$ $= \frac{x^{-\frac{1}{2}}(2x(6x - 2) - (3x^2 - 2x + 1))}{2x}$ $= \frac{12x^2 - 4x - 3x^2 + 2x - 1}{2x\sqrt{x}}$ $= \frac{9x^2 - 2x - 1}{2x\sqrt{x}}$	2	1 mark correct differentiation 1 mark correct simplification (negative indices acceptable) 1 mark correct use of quotient rule 1 mark to this factorisation	
ii)	$y = \frac{x^2}{1 - x^3}$ $u = x^2 \quad v = 1 - x^3$ $u' = 2x \quad v' = -3x^2$ $y' = \frac{vu' - uv'}{v^2}$ $= \frac{(1 - x^3) \times 2x - x^2 \times -3x^2}{(1 - x^3)^2}$ $= \frac{2x - 2x^4 + 3x^4}{(1 - x^3)^2}$ $= \frac{x^4 + 2x}{(1 - x^3)^2}$ $= \frac{x(x^3 + 2)}{(1 - x^3)^2}$	2	1 mark correct use of quotient rule 1 mark correct factorised form	

b)iii)	$y = (x+1) \sqrt{4x-1}$ $u = (x+1) \quad v = (4x-1)^{\frac{1}{2}}$ $u' = 1 \quad v' = \frac{1}{2} \times 4(4x-1)^{-\frac{1}{2}}$ $= 2(4x-1)^{-\frac{1}{2}}$ $y' = uv' + vu'$ $= (x+1) \times 2(4x-1)^{-\frac{1}{2}} + (4x-1)^{\frac{1}{2}} \times 1$ $= (4x-1)^{-\frac{1}{2}} [2(x+1) + (4x-1)]$ $= \frac{(2x+2+4x-1)}{\sqrt{4x-1}}$ $= \frac{6x+1}{\sqrt{4x-1}}$	3	1 mark correct use of product rule 1 mark correct progress in factorisation 1 mark correct fully factorised
c)	$y = \frac{a}{x+2} = a(x+2)^{-1}$ $y' = a \times -1(x+2)^{-2}$ $= \frac{-a}{(x+2)^2}$ $x = 2, m = \frac{1}{4}$ $\frac{1}{4} = \frac{-a}{(x+2)^2} = \frac{-a}{(2+2)^2}$ $= \frac{-a}{16}$ $16 = -4a$ $a = -4$	2	1 mark for expression equating correct derivative to given gradient 1 mark correct
d)	$x^2 + bx + c = 0$ Roots are $\alpha, 3\alpha$ Sum: $\alpha + 3\alpha = -b$ $4\alpha = -b$ Product: $\alpha \times 3\alpha = c$ $3\alpha^2 = c$ From sum, $\alpha = -\frac{b}{4}$ $3\alpha^2 = 3\left(-\frac{b}{4}\right)^2$ So from product: $\frac{3b^2}{16} = c$ $3b^2 - 16 = 0$	2	1 mark correct expression for both sum and product of roots 1 mark correct derivation of result

Question 13		Preliminary Mathematics Extension 1	2012	
Part	Solution	Marks	Comment	
(a)	$\frac{\cos \theta}{1 - \sin \theta} - \frac{1 - t}{1 + t}$ $LHS = \frac{\frac{1 - t^2}{1 + t^2}}{1 - \frac{2t}{1 + t^2}} \quad u \sin \theta \cos \theta = \frac{1 - t^2}{1 + t^2} \quad \text{and} \quad \sin \theta = \frac{2t}{1 + t^2} \quad \text{where} \quad \tan \frac{\theta}{2} = t$ $= \frac{1 - t^2}{1 + t^2 - 2t}$ $= \frac{1 - t^2}{(1 - t)^2}$ $= \frac{(1 + t)(1 - t)}{(1 - t)^2}$ $= \frac{(1 + t)}{(1 - t)}$ $= RHS$	3	1 for correct substitution of correct t results 1 for correct simplification 1 for correct simplification to reach result	
(b)	$\sqrt{3} \sin x - \cos x = R \sin(x - \alpha)$ $= R \sin x \cos \alpha - R \cos x \sin \alpha$ Equating coefficients: $R \cos \alpha = \sqrt{3} \quad R \sin \alpha = 1$ For R: $R^2 \cos^2 \alpha + R^2 \sin^2 \alpha = (\sqrt{3})^2 + 1$ $R^2 (\cos^2 \alpha + \sin^2 \alpha) = 4$ $R^2 = 4$ $R = 2, \quad R > 0$ $\tan \alpha = \frac{\sin \alpha}{\cos \alpha}$ For α : $= \frac{1}{\sqrt{3}}$ $\alpha = 30^\circ \quad 0^\circ \leq \alpha \leq 90^\circ$ So: $\sqrt{3} \sin x - \cos x = 2 \sin(x - 30^\circ)$	2	May use long or short method 1 mark each correct R and α .	

Question 13		Preliminary Mathematics Extension 1	2012	
Part	Solution	Marks	Comment	
b)ii)	$\sqrt{3} \sin x - \cos x = 1 \quad 0^\circ \leq x \leq 360^\circ$ $2 \sin(x - 30^\circ) = 1 \quad \text{above } -30^\circ \leq x \leq 330^\circ$ $\sin(x - 30^\circ) = \frac{1}{2} \quad \text{Quadrants 1 \& 2}$ $(x - 30^\circ) = 30^\circ, 150^\circ$ $x = 60^\circ, 180^\circ$	2	1 mark 1 mark BOTH solutions	
(c)(i)	$\frac{1 - \cos 2\theta}{1 + \cos 2\theta} = \frac{1 - (1 - 2\sin^2 \theta)}{1 + (2\cos^2 \theta - 1)}$ $= \frac{2\sin^2 \theta}{2\cos^2 \theta}$ $= \tan^2 \theta$	2	1 1 Any correct substitution	
(ii)	$\tan^2 22 \frac{1}{2}^\circ = \frac{1 - \cos 45^\circ}{1 + \cos 45^\circ}$ $= \frac{1 - \frac{1}{\sqrt{2}}}{1 + \frac{1}{\sqrt{2}}}$ $= \frac{\frac{\sqrt{2} - 1}{\sqrt{2}}}{\frac{\sqrt{2} + 1}{\sqrt{2}}}$ $= \frac{\sqrt{2} - 1}{\sqrt{2} + 1} \times \frac{\sqrt{2} - 1}{\sqrt{2} - 1}$ $\tan^2 22 \frac{1}{2}^\circ = \frac{(\sqrt{2} - 1)^2}{2 - 1}$ $= (\sqrt{2} - 1)^2$ $\therefore \tan 22 \frac{1}{2}^\circ = \sqrt{2} - 1$	2	1 using result above correctly 	

Question 13		Preliminary Mathematics Extension 1	2012	
Part	Solution	Marks	Comment	
(d)(i)	$x^4 + 4 \equiv (x^2 + a)^2 - (bx)^2$ $\equiv x^4 + (2a - b^2)x^2 + a^2$ <i>equating coefficients</i> $2a - b^2 = 0 \Rightarrow b^2 = 2a \quad (\text{since coefficient of } x^2 \text{ is } 0)$ $a^2 = 4 \Rightarrow a = \pm 2 \quad (\text{since } 4 \text{ is the constant term})$ <i>but } b^2 = 2a, \text{ ie } a = 2 \text{ since } b^2 > 0</i> $\text{so } b^2 = 4 \Rightarrow b = \pm 2$	2	1 for correct expansion and grouping of coefficients according to powers of x 1 for a & b	
(ii)	$x^4 + 4 \equiv (x^2 + 2)^2 - (2x)^2$ $= \{(x^2 + 2) + 2x\} \{(x^2 + 2) - 2x\}$ $= (x^2 + 2x + 2)(x^2 - 2x + 2)$	2	1 correct factorisation into difference of squares 1 correct result	